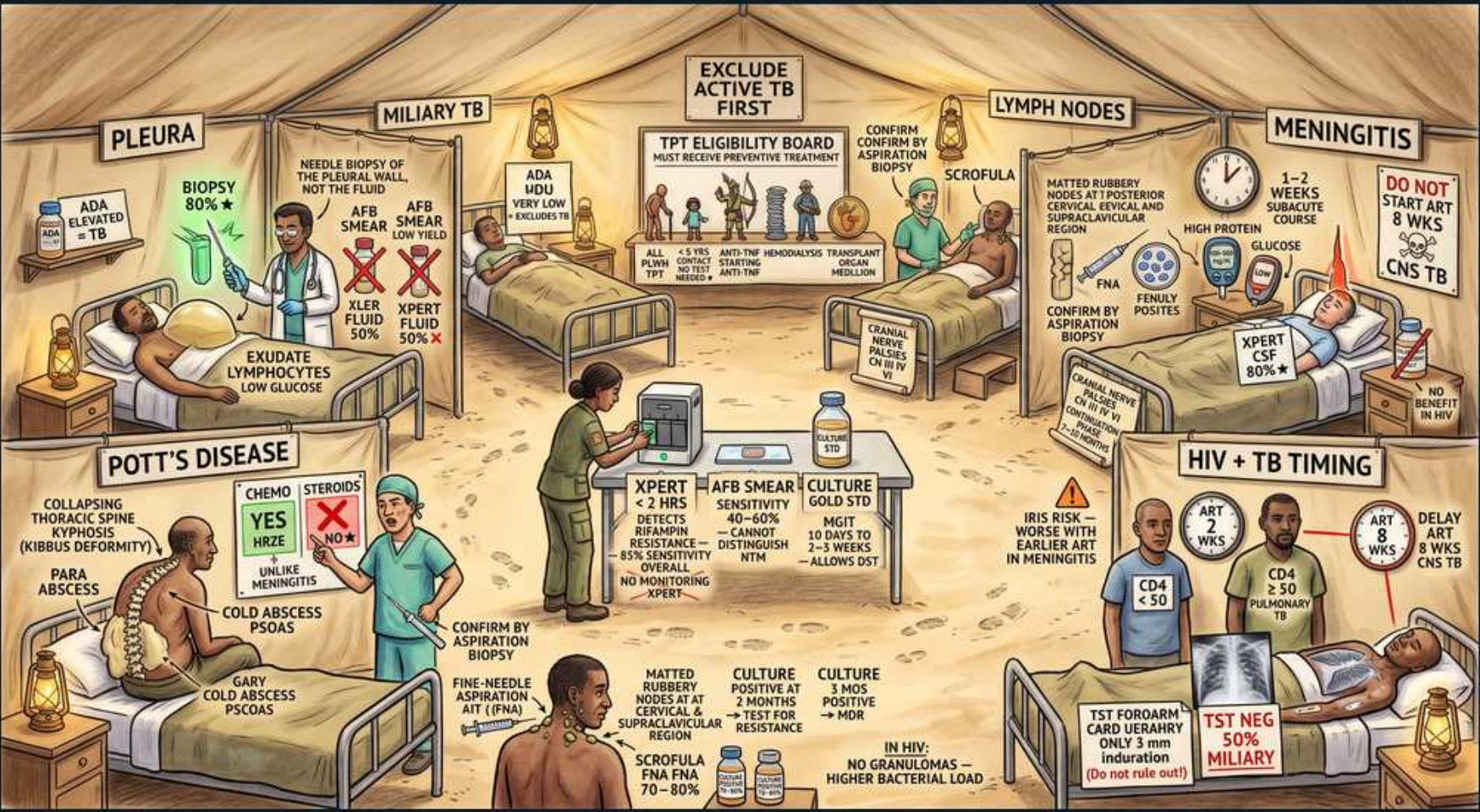


Extrapulmonary TB — Sites, Fluid Analysis & HIV Timing



1. Pleural TB — fluid vs biopsy

WHAT YOU SEE

'PLEURA', 'ADA ELEVATED = TB', 'BIOPSY 80%'

WHAT IT ENCODES

TB pleuritis gives an exudative, lymphocyte-predominant effusion with HIGH adenosine deaminase (ADA) and low glucose. Pleural fluid AFB/culture yield is low (~30–50%); pleural BIOPSY yields ~80%.

BOARD TRAP

Diagnose pleural TB by biopsy of the pleura, NOT by fluid culture — fluid smear is usually negative.

2. Pleural fluid: exudate, low glucose

WHAT YOU SEE

'EXUDATE LYMPHOCYTES LOW GLUCOSE', 'AFB SMEAR LOW YIELD'

WHAT IT ENCODES

Lymphocytic exudate with elevated ADA (>40) and low glucose is the classic TB pleural profile; mesothelial cells are characteristically scarce.

BOARD TRAP

Absence of mesothelial cells supports TB; eosinophilic effusion argues AGAINST TB.

3. Miliary TB — disseminated

WHAT YOU SEE

'MILIARY TB', 'ADA WBC VERY LOW = EXCLUDES TB'

WHAT IT ENCODES

Miliary TB = hematogenous dissemination with countless 1–2 mm nodules across organs. High mortality; can cause ARDS and marrow involvement. Common in HIV and young children.

BOARD TRAP

In a CSF/effusion, a very low ADA argues against TB; miliary disease itself needs urgent multi-drug therapy + biopsy of an accessible site.

4. Lymph node TB — scrofula

WHAT YOU SEE

'LYMPH NODES', 'SCROFULA', 'MATTED RUBBERY NODES'

WHAT IT ENCODES

Scrofula = TB cervical lymphadenitis, classically matted, rubbery posterior cervical/supraclavicular nodes. Most common extrapulmonary site overall.

BOARD TRAP

Confirm by EXCISIONAL/aspiration biopsy and culture; FNA can miss it. In children, nontuberculous mycobacteria can mimic scrofula.

5. TB meningitis — basilar

WHAT YOU SEE

'MENINGITIS', '1–2 WEEKS SUBACUTE COURSE', 'HIGH PROTEIN GLUCOSE'

WHAT IT ENCODES

TB meningitis is subacute (1–2 weeks) with high CSF protein, LOW glucose, lymphocytic pleocytosis, and basilar meningitis affecting cranial nerves. Xpert/CSF AFB low sensitivity.

BOARD TRAP

Add CORTICOSTEROIDS to TB meningitis and pericarditis to reduce mortality/complications — a high-yield exception to 'no steroids in infection'.

6. Xpert CSF ~80% / culture gold

WHAT YOU SEE

'XPERT CSF 80%', 'CULTURE GOLD STD'

WHAT IT ENCODES

GeneXpert MTB/RIF is rapid (<2 h) and also detects rifampin resistance; sensitivity ~40–65% on paucibacillary samples. Culture (MGIT) remains the gold standard and allows full DST.

BOARD TRAP

A negative Xpert does NOT rule out paucibacillary TB (meningitis, pleura) — culture and biopsy still needed.

7. Pott disease — spinal TB

WHAT YOU SEE

'POTT'S DISEASE', 'COLLAPSING THORACIC SPINE KYPHOSIS (gibbus)', 'COLD ABSCESS PSOAS'

WHAT IT ENCODES

Spinal TB (Pott disease) favors the lower thoracic spine causing anterior vertebral collapse → gibbus kyphosis, and cold paraspinal/psoas abscesses. Treat with HRZE; surgery for cord compression.

BOARD TRAP

A 'cold' (non-tender, afebrile) psoas abscess tracking from the spine = TB until proven otherwise; pyogenic abscesses are hot/painful.

8. Chemo yes, steroids no (Pott)

WHAT YOU SEE

'CHEMO YES HRZE', 'STERIODS NO'

WHAT IT ENCODES

Standard anti-TB chemotherapy (HRZE) treats Pott disease; steroids are NOT routinely indicated for skeletal TB (unlike CNS/pericardial TB).

BOARD TRAP

Steroids help TB meningitis/pericarditis but NOT spinal/pleural TB — match the steroid indication to the site.

9. Xpert <2 h, RIF resistance

WHAT YOU SEE

'XPERT <2 HRS DETECTS RIFAMPIN RESISTANCE 85% SENSITIVE'

WHAT IT ENCODES

GeneXpert gives a result in under 2 hours and simultaneously screens for rifampin resistance, a surrogate for MDR-TB.

BOARD TRAP

Rifampin resistance on Xpert flags possible MDR-TB — escalate to second-line regimen and full DST.

10. AFB smear sensitivity 40–65%

WHAT YOU SEE

'AFB SMEAR SENSITIVITY 40–65% CANNOT DISTINGUISH NTM'

WHAT IT ENCODES

Sputum AFB smear is fast and cheap but only 40–65% sensitive and cannot distinguish *M. tuberculosis* from nontuberculous mycobacteria.

BOARD TRAP

A positive smear ≠ confirmed TB (could be NTM); a negative smear does not exclude TB — culture/NAAT confirm.

11. Culture gold standard, slow

WHAT YOU SEE

'CULTURE GOLD STD', 'MGIT 1–2 WEEKS', 'allows DST'

WHAT IT ENCODES

Liquid MGIT culture is the gold standard, turning positive in ~1–2 weeks (solid Löwenstein–Jensen up to 8 weeks) and enabling drug-susceptibility testing.

BOARD TRAP

Culture confirms diagnosis and resistance but is too slow to guide initial therapy — start empiric RIPE on suspicion.

12. TPT eligibility — treat LTBI

WHAT YOU SEE

'EXCLUDE ACTIVE TB FIRST', 'TPT ELIGIBILITY BOARD must receive preventive treatment'

WHAT IT ENCODES

High-risk LTBI (HIV, transplant, anti-TNF candidates, recent converters) should receive TB preventive therapy after active TB is excluded.

BOARD TRAP

Always exclude active disease before LTBI monotherapy to avoid generating resistance.

13. HIV: start ART by CD4 timing

WHAT YOU SEE

'HIV + TB TIMING', 'CD4 <50 → ART 2 WKS', 'CD4 >50 → ART 8 WKS'

WHAT IT ENCODES

In HIV/TB co-infection, start TB therapy first, then ART: within 2 weeks if CD4 <50, by 8 weeks if CD4 ≥50 — to limit mortality while reducing IRIS.

BOARD TRAP

TB MENINGITIS is the exception — DELAY ART ~8 weeks regardless of CD4 because early ART worsens CNS IRIS.

14. Delay ART 8 wks in CNS TB

WHAT YOU SEE

'DELAY ART 8 WKS CNS TB', 'IRIS RISK — WORSE WITH EARLIER ART IN MENINGITIS'

WHAT IT ENCODES

Immune reconstitution inflammatory syndrome (IRIS) after ART is dangerous in CNS TB; delaying ART to ~8 weeks reduces fatal CNS IRIS.

BOARD TRAP

Don't rush ART in TB meningitis even with very low CD4 — opposite of the usual 'low CD4 = start ART early' rule.

15. HIV TB: paucibacillary, no granulomas

WHAT YOU SEE

'IN HIV: NO GRANULOMAS — HIGHER BACTERIAL LOAD', 'TST NEG 50% MILIARY'

WHAT IT ENCODES

Advanced HIV impairs granuloma formation, giving disseminated/miliary disease with higher organism load but frequently anergic (false-negative) TST.

BOARD TRAP

A negative TST does NOT exclude TB in advanced HIV — anergy is common; rely on IGRA, NAAT, culture, and clinical suspicion.